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Database Systems

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CRITICAL EVALUATION (06 Marks)

1.1 Strength

The EERD accurately depicts WAREHOUSE_WORKER and DRIVER as subsets of the EMPLOYEE entity. This appropriately reflects the company policy that Foodz@Yourz uses two primary staff categories, each with unique characteristics (e.g., shift for warehouse workers and licence number for drivers). This design adheres to the proper inheritance principle of conceptual modelling.

The diagram contains key components that are directly related to Foodz@Yourz's business operations, including ORDER, PAYMENT, DELIVERY, STORE, and PRODUCT. This proves the AI's accurate comprehension of the primary business operations of grocery delivery, payment, and ordering.

The many-to-many relationships between STORE-PRODUCT and BASKET-PRODUCT are successfully handled by the addition of STORE_PRODUCT and BASKET_ITEM. Because it guarantees accurate representation of product availability and member basket composition, this is a suitable conceptual design choice.

1.2 Weakness

Entities like ORDER_ASSIGNMENT and COLLECTION_ORDER, which add implementation-level detail not necessary in a conceptual design, are included in the EERD. Business rules and data relationships, not internal processing logic, should be the main focus of a conceptual EERD.

Payment and Vehicle Specialisation Missed: The EERD does not use specialisation to model PAYMENT (card/mobile) and VEHICLE (two/four-wheeler). This goes against EER principles, which state that subtypes must have unique attributes (e.g., card details for card payment). Furthermore, the business constraint that a two-wheeler can only handle small collections cannot be enforced due to the lack of Vehicle and Collection specialisation.

Clear multiplicities are absent from some relationships, such as those between ORDER and PAYMENT or MEMBER and ORDER. For example, each member may have multiple orders, but each order should only require a single payment. Inaccurate or missing multiplicities impair the model's precision and interpretability.

DECISION LOG

1. Model Employee as a supertype with Driver and WarehouseWorker subtypes (specialisation) is the first decision.

Justification: The Foodz@Yourz brief distinguishes between drivers and warehouse employees based on their roles and characteristics (for example, drivers have license_no, warehouse workers have shift). A common conceptual modelling best practice is to use specialisation to make role-specific attributes explicit while preserving common attributes (employee_id, name) at the supertype and avoiding duplication.

2. Use vehicle_type (two-wheeler or four-wheeler) and capacity attribute (or subtypes) to represent the vehicle and record the capacity constraint.

Justification: Since two-wheelers can only manage small collections, the brief specifies different operating limits for them than for four-wheelers. This business rule is conceptually captured and made explicit for implementation by adding TwoWheeler/FourWheeler subtypes.

3. Model Product as a Supertype with Perishable and NonPerishable Considerations . Focus on Food and Drink with SoftDrink and Alcoholic Drink subtypes.

Justification: According to the brief, drinks must have certain characteristics (ml_per_serving , legal_min_age) and perishable goods must have expiration dates. For various product families, multi-level specialisation clarifies domain constraints and lowers NULLs.

4. Model Address and Member as distinct entities, with Address associated with Member

Justification: Orders must maintain the delivery address used at the time of order, even though members may have multiple delivery addresses. When members update their profiles, decoupling addresses keeps past delivery data intact.

5. Model Payment as an entity connected to Order capturing method, amount, and paid at.

Justification: Card and mobile payments are accepted by the system. Separately recording payment information (and connecting it to orders with a distinct multiplicity) maintains transactional integrity and complies with the brief's requirements for reconciliation and audit.

ENTITIES

Entity Name	Brief Explanation
w2149496_Product	Represents the product offered by Foodz@Yourz. Each product is offered by one or more store
w2149496_Store	Represents a location/outlet where products are sold or stored.
w2149496_Collection	Represents a set of possibly deliveries.
w2149496_Perishable	A subclass of Product that has an expiration date.
w2149496_Non_Perishable	A subclass of Product that doesn't have an expiration date.
w2149496_Food	A subclass of Perishable product
w2149496_Drink	A subclass of Perishable product
w2149496_Soft_Drink	A subclass of Non Perishable product
w2149496_Alcoholic_Bev	A subclass of Non Perishable product
w2149496_Member	Represents a registered customer.
w2149496_Basket	Represents a shopping cart created by a member.
w2149496_Payment	Represents a payment transaction for a basket.
w2149496_Card_Pay	A subclass of payment using a card
w2149496_Mobile_Pay	A subclass of payment using mobile platform
w2149496_Small_Coll	A subclass of Collection with a maximum order limit.

w2149496_Large_Coll	A subclass of Collection with a minimum order limit.
w2149496_Order	Represents the order confirmed by customer
w2149496_Delivery	Represents the delivery details of the customer who placed the order
w2149496_Address	Represents the home address of the customer
w2149496_Employee	Represents a person employed by Foodz@Yourz.
w2149496_Driver	A subclass of employee who drives vehicle
w2149496_Warehouse_Worker	A subclass of Employee who works in the warehouse/store.
w2149496_Vehicle	Represents the vehicle used for deliveries
w2149496_Two_W	Subclass of vehicle used to deliver the small collection of orders
w2149496_Four_W	Subclass of vehicle used to deliver the large collection of orders

SPECIALISATIONS

General Entity	Specialised entity	Brief explanation
w2149496_Product	w2149496_Perishable, w2149496_Non Perishable	Products are grouped according to whether or not they expire. "Best before" and "use by" dates are necessary for perishable goods but not for non-perishable ones.
w2149496_Perishable	w2149496_Food, w2149496_Drink	Food with dietary and allergen tags and drink with serving volume attributes are two more categories for perishable goods.
w2149496_Non Perishable	w2149496_Soft_Drink, w2149496_Alcoholic_beve	Soft drinks containing sugar and alcoholic beverages including alcohol % and legal age limits are

		two categories for non-perishable goods.
w2149496_Collection	w2149496_Small_Coll, w2149496_Large_Coll	Size restrictions are used to categorise collections. There is a minimum restriction for large collections and a maximum limit for small collections.
w2149496_Payment	w2149496_Card_Pay, w2149496_Mobile_Pay	Payments are categorised according to the settlement method. Whereas mobile phone Pay records the mobile phone number, Card Pay records the card data.
w2149496_Vehicle	w2149496_Two_W, w2149496_Four_W	vehicles are classed according to the number of wheels they have.
w2149496_Employee	w2149496_Driver, w2149496_Warehouse_Worker	Workers are identified by their main function. While warehouse employees have a set shift, drivers use vehicles and have a licence number.

RELATIONSHIPS & MULTIPLICITIES

Entity Name	Multiplicity	Relationship	Multiplicity	Entity name	Brief justifications for the multiplicity
w2149496_Member	1..*	creates	0..1	w2149496_Basket	1. Each Member may create one or more Baskets. 2. Each Basket must be created by exactly one Member. 3. A Basket cannot exist without being linked to a Member, showing mandatory participation on the Basket side. 4. The Member side is optional because a member may register but not yet create a basket.
w2149496_Basket	1..1	based on	1..1	w2149496_Order	1. Each Order must be based on one Basket. 2. Each Basket can result in only one Order. 3. The relationship is mandatory for both entities, as an Order cannot exist without a Basket. 4. This ensures referential integrity between shopping and ordering processes.
w2149496_Order	1..1	can have	1..1	w2149496_Delivery	1. Each Order must have one corresponding Delivery. 2. Each Delivery relates to exactly one Order.

					3. This relationship is mandatory for both entities, as every order must be delivered. 4. The one-to-one relationship ensures tracking of delivery per order.
w2149496_Store	1..*	has	1..*	w2149496_Product	1. Each Store may hold multiple Products. 2. Each Product may be stocked in more than one Store. 3. The relationship is many-to-many, reflecting inventory distribution across branches. 4. Both sides have mandatory participation to ensure product availability.
w2149496_Delivery	1..1	used for	0..*	w2149496_Address	1. Each Delivery must be associated with exactly one Address (1..1). 2. An Address may be used for zero or many Deliveries (0..*), e.g., repeat customers or unused saved addresses. 3. Addresses are shared resources across deliveries, so the address side is many/optional. 4. The mandatory side is Delivery, because a delivery must have a destination
w2149496_Order	1..1	settled by	1..1	w2149496_Payment	1. Each Order is settled by exactly one Payment transaction (1..1). 2. Each Payment record belongs to exactly one Order (1..1). 3. The diagram shows this as mandatory on both sides to ensure a completed payment is recorded for every order.

					4. This provides accountability and an auditable payment per order.
w2149496_Payment	1..1	specialisation of	{Mandatory, Or}	w2149496_Card_Pay / w2149496_Mobile_Pay	1. Payment must be one of the specialised methods (Card or Mobile) {Mandatory, Or} as in the diagram. 2. Exactly one specific payment subtype (Card_Pay or Mobile_Pay) records method-specific attributes. 3. The mandatory or constraint ensures a payment record cannot exist without specifying its type. 4. This models alternative method types with required choice of one subtype.
w2149496_Product	{Mandatory, Or}	specialised into	{Mandatory, Or}	w2149496_Perisable / w2149496_Non_Perisable	1. Product is specialised into Perisable or Non_Perisable 2. The diagram shows a mandatory choice (product must be classified) {Mandatory, Or} 3. Within Perisable, further subtypes (Food/Drink) exist, and some specialisations are {Mandatory, And} where attributes from both may apply (diagram markings). 4. This enforces correct product-type attributes (dates, use_by etc.) only when applicable.
w2149496_Drink	{Mandatory, And}	specialised into	1..*	w2149496_Soft_Drink / w2149496_Alcoholic_bev	1. Drink may be further specialised into Soft_Drink and Alcoholic_bev (diagram shows {Mandatory, And}, meaning drink-type attributes may require additional subtype attributes). 2. Alcoholic_bev includes

					percentage and legal_min_age attributes. 3. Soft_Drink includes sugar_100ml attribute. 4. The specialisation enforces the presence of correct extra attributes depending on the subtype.
w2149496_Collection	1..*	can have	1..1	w2149496_Order	1. Each Collection (Small or Large) can include one or more Orders (1..*). 2. Each Order must belong to exactly one Collection (1..1), ensuring grouping for delivery or promotion. 3. The relationship is mandatory for Order, as shown by the solid line (each order must be part of a collection). 4. The Collection side allows multiple orders, enabling batch processing or grouped dispatches.
w2149496_Two_W	1..1	can have	1..*	w2149496_Small_Collection	1. Each Two_W vehicle can handle multiple Small_Collections (1..*). 2. Each Small_Collection is assigned to exactly one Two_W vehicle (1..1). 3. This represents capacity and practicality — small vehicles can deliver smaller bundles. 4. The relationship is mandatory for Small_Collection, as shown by the solid connector in the EERD.
w2149496_Four_W	1..1	assigned	1..*	w2149496_Large_Collection	1. Each Four_W (four-wheeler) can assign deliveries to multiple Large_Collections (1..*).

					2. Each Large_Collection is assigned to exactly one Four_W vehicle (1..1). 3. The relationship represents large delivery operations needing higher capacity vehicles. 4. It's mandatory for Large_Collection, ensuring proper vehicle assignment for transportation.
w2149496_Driver	1..1	uses	1..*	w2149496_Vehicle	1. Each Driver may use one or more Vehicles across different shifts (1..*). 2. Each Vehicle is used by exactly one Driver at a time (1..1). 3. This ensures operational accountability for each delivery run. 4. The relationship is mandatory for Vehicle, since no vehicle can operate without a driver.
w2149496_Basket	1..*	contains	1..*	w2149496_Product	1. Each Basket can contain multiple Products (1..). 2. Each Product can appear in multiple Baskets (1..). 3. This many-to-many relationship captures the customer's selection of several products per basket. 4. It is mandatory for both entities to represent valid basket product combinations in the ordering process.

ATTRIBUTES & PRIMARY KEYS

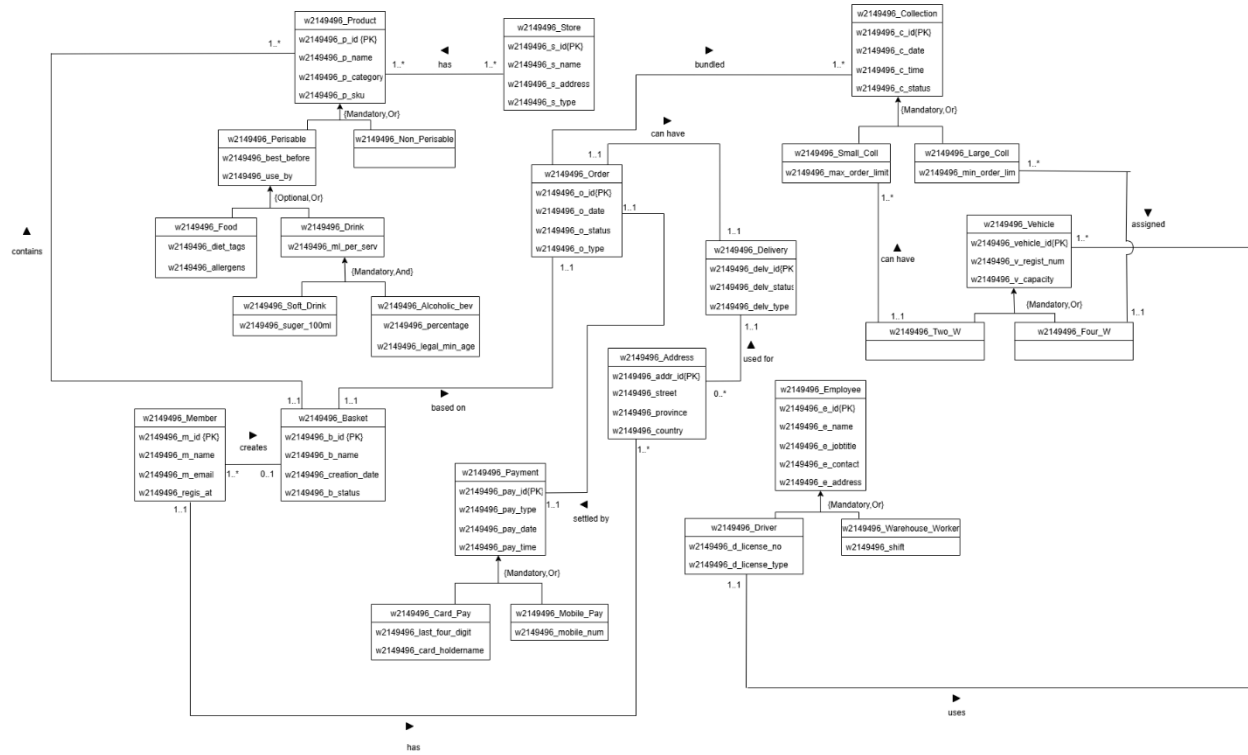
Entity Name	Attributes for each entity (include PK)	Brief explanation
w2149496_Product	w2149496_p_id (PK), w2149496_p_name, w2149496_p_category, w2149496_p_sku	The product's unique identifier (PK), name, description, and stock-keeping unit are some of its most important characteristics.
w2149496_Perishable	w2149496_best_before, w2149496_use_by	Unique characteristics for perishable goods, including required "best before" dates and flexible "use by" dates.
w2149496_Non_Perishable	(No any unique attributes)	Represents the Non Perishable food category Inherits all attributes from Product
w2149496_Food	w2149496_diet_tags, w2149496_allergens	Details about food things, such as allergen warnings and dietary restrictions (tags).
w2149496_Drink	w2149496_ml_per_serv	A specific property for drinks that measures the amount of liquid in a serving in millilitres.
w2149496_Soft_Drink	w2149496_suger_100ml	This particular soft drink attribute describes the amount of sugar in 100 milliliters.

w2149496_Alcoholic_bev	w2149496_percentage, w2149496_legal_min_age	describing the minimum legal buying age and the proportion of alcohol
w2149496_Store	w2149496_s_id (PK), w2149496_s_name, w2149496_s_address, w2149496_s_type	characteristics that define the store or outlet, such as its name, physical address, kind (e.g., small, large), and unique identification (PK).
w2149496_Member —	w2149496_m_id (PK), w2149496_m_name, w2149496_m_email, w2149496_regis_at	Features for consumers who have registered, such as their name, login timestamp, email address, and unique identification (PK).
w2149496_Basket	w2149496_b_id (PK), w2149496_b_name, w2149496_creation_date, w2149496_b_status	characteristics that keep track of the shopping basket, such as its name, creation date, unique identifier (PK), and current status (e.g., active, finalised).
w2149496_Payment	w2149496_pay_id (PK), w2149496_pay_type, w2149496_pay_date, w2149496_pay_time	features that track payment transactions, such as the date, type (credit, mobile, etc.), unique identification (PK), and transaction time.
w2149496_Card_Pay	w2149496_last_four_digit, w2149496_card_holdername	particular characteristics for card payments, including the last four digits and the cardholder's name for security.
w2149496_Mobile_Pay	w2149496_mobile_num	Recording the mobile phone number used for the transaction is

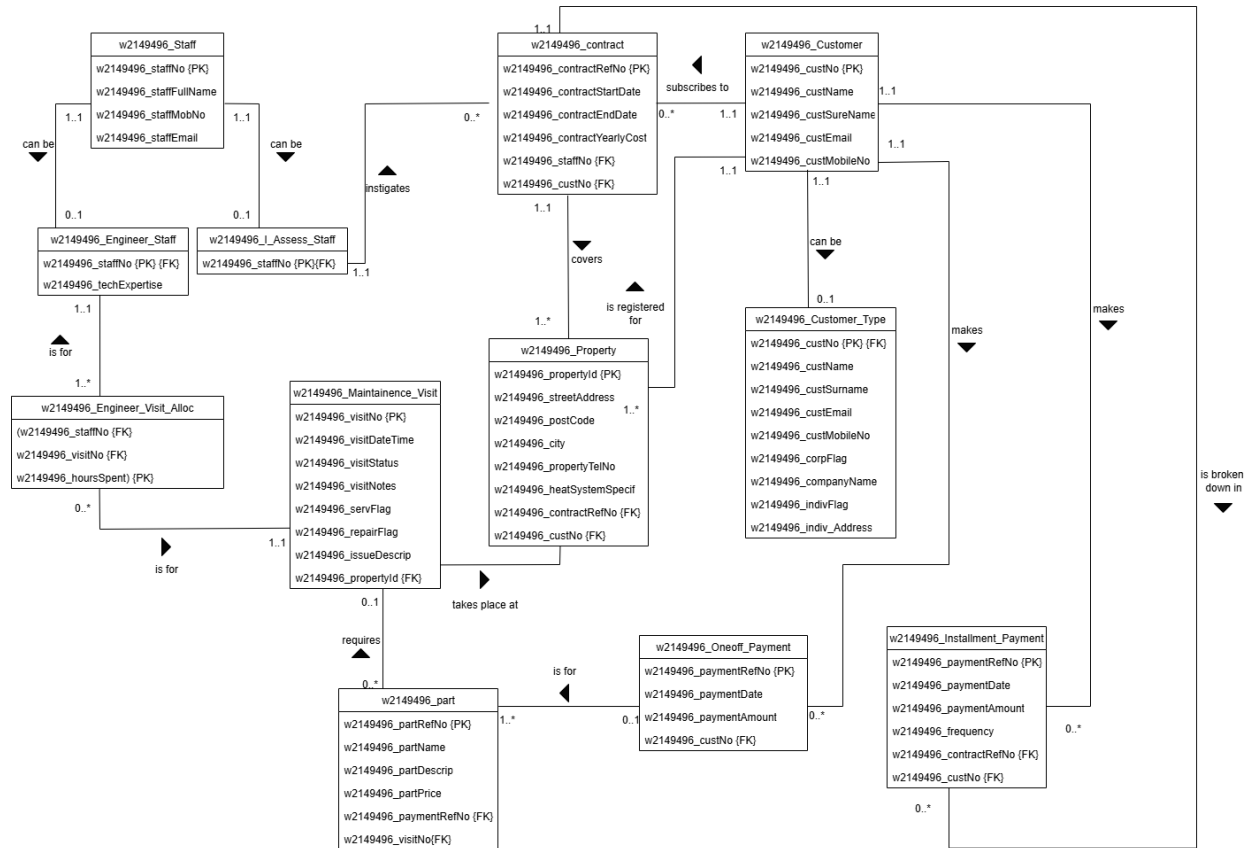
		a unique feature for mobile payments.
w2149496_Order	w2149496_o_id (PK), w2149496_o_date, w2149496_o_status, w2149496_o_type	characteristics of a verified customer order, such as its date, status, type (e.g., standard, express), and unique identifier (PK).
w2149496_Delivery	w2149496_delv_id (PK), w2149496_delv_status, w2149496_delv_type	Attributes tracking the shipment, including its unique identifier (PK), current status, and type (e.g., standard, scheduled).
w2149496_Address	w2149496_addr_id (PK), w2149496_street, w2149496_province, w2149496_country	characteristics that define a physical location, such as its country, province, street details, and unique identification (PK).
w2149496_Employee	w2149496_e_id (PK), w2149496_e_name, w2149496_e_jobtitle, w2149496_e_contact, w2149496_e_address	characteristics that define an employee, such as their name, work title, contact details, primary address, and unique identification (PK).
w2149496_Driver	w2149496_d_license_no, w2149496_d_license_type	particular characteristics for workers in the driver position, including their licence type and unique licence number.
w2149496_Warehouse_Worker	w2149496_shift	particular characteristic for warehouse workers that describes their designated work shift.

w2149496_Vehicle	w2149496_v_id (PK), w2149496_v_regist_num, w2149496_v_capacity	characteristics that define a vehicle, such as its capacity (such as volume or weight), registration number, and unique identification (PK).
w2149496_Two_W	(No unique attributes)	represents the category of two-wheeled vehicles. inherits every characteristic of the vehicle.
w2149496_Four_W	(No unique attributes)	represents the category of four-wheeled vehicles. inherits every characteristic of the vehicle.
w2149496_Collection	w2149496_c_id (PK), w2149496_c_date, w2149496_c_time, w2149496_c_status	characteristics that keep track of the collection or fulfilment event, such as its date, time, and current state, as well as its unique identifier (PK).
w2149496_Small_Coll	w2149496_max_order_limit	particular feature that specifies the highest value or size permitted per order for tiny collections.
w2149496_Large_Coll	w2149496_min_order_lim	particular feature that specifies the minimum size or value needed for each order in huge collections.

QUALITY & NOTATIONS OF CONCEPTUAL EERD



MAPPING CONCEPTUAL EERD TO LOGICAL ERD



MAPPING LOG

- 1) All of the entities and relationships from the Conceptual EERD were resolved using a methodical Logical Mapping Strategy when creating the Logical ERD for BoilHeaterz. Every powerful entity and attributes was converted into a table, with names prefixed by *w2149496_*.
- 2) The table-per-subtype method was used to map subtype entities like Engineer, Initial_Assessor, Corporate, and Individual. In order to preserve one-to-one relationships, each subtype inherited the primary key of its supertype as both a primary key and a foreign key.
- 3) By positioning the foreign key on the "many" side (for example, *w2149496_custNo* in *w2149496_Contract*), one-to-many relationships were established. Using the associative table *w2149496_Engineer_Visit_Alloc* and composite keys (*w2149496_staffNo*, *w2149496_visitNo*), the many-to-many relationship between Engineer and Maintenance_Visit was resolved. Similarly, Maintenance_Visit and Part were linked by *w2149496_Servicing* and *w2149496_Repair*.
- 4) The final UML-based Logical ERD ensured referential integrity and logical consistency by introducing all foreign keys in accordance with cardinality and participation rules.